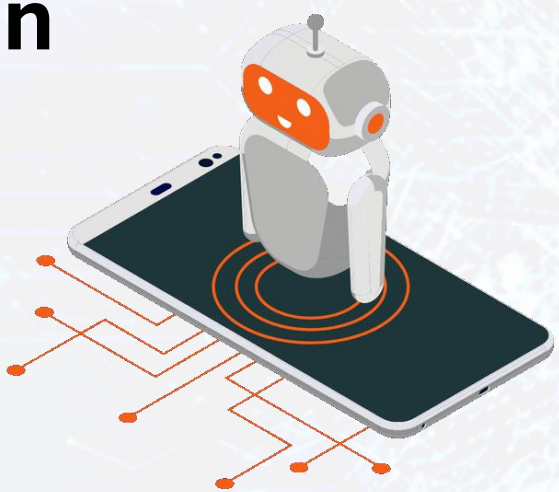
សាកលវិទ្យាល័យភ្នំពេញអន្តរជាតិ
PHNOM PENH INTERNATIONAL UNIVERSITY

Student Performance Prediction

Machine Learning Course

Regression model for final grade prediction



Lecture: **HIM Sok Long**

Present by

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Our Content

- Introduction
- Project overview
- Problem statement
- Dataset description
- Data Preprocessing
- EDA (Exploratory Data Analysis)
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- Methodology
- Algorithms Used
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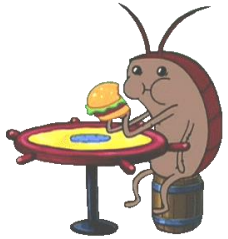
  - Random Forest Regressor
  - Gradient Boosting Regressor
- Evaluation Metrics
- Model Comparison Table
- Visual Comparison
- Best Model Analysis
- Challenges
- Conclusion

Mr. Kong Sinh

@kongsinh-dev @kong_sinh @mr.kongsinh

I am honored to present

- Introduction
- Project overview
- Problem statement
- Dataset description
- Gradient Boosting Algorithm



Var. Som Soknouch

@sokncouch @soknouchsom

I'll show you about

- Data Preprocessing
- EDA (Exploratory Data Analysis)
- Key Data Insights
- Methodology
- Random Forest Regressor

Ms. Hoeurng Phally

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I'm going to tell you

- Linear Regression
- Evaluation Metrics
- Model Comparison Table
- Best Model Analysis
- Challenges
- Conclusion

Introduction

Student performance is an important topic in education because it helps schools understand how students learn and what factors may affect their academic results.

In this project, we use **machine learning** to predict a student's final grade based on different types of information, such as study time, absences, previous grades, family background, and school support.

The main goal of this project is not to judge students, but to help identify students who may need support earlier. By using data, schools and teachers can better understand student learning patterns and make better academic decisions.

This project is a **regression project** because the model predicts a numerical value, which is the student's final grade, called **G3**.

01 Main idea

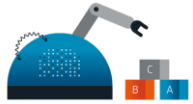
Use past student information to estimate final grade G3.

02 Machine learning value

Learns hidden relationships from many student factors.

03 Academic purpose

Supports early decision-making and model comparison.

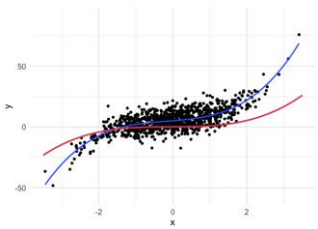


Project Overview

The purpose of this project is to predict student academic performance based on different student-related factors like **demographic**, **academic**, **family**, and **school-related** features.

WHY IT MATTERS

- Helps teachers identify academic risk earlier.
- Shows which factors are related to student outcomes.
- Supports data-driven academic intervention.
- Provides practical experience with ML workflow.



Input

Student features such as G1, G2, study time, failures, absences, and support.

Model

Regression algorithms learn patterns from training data.

Output

Predicted final grade G3.

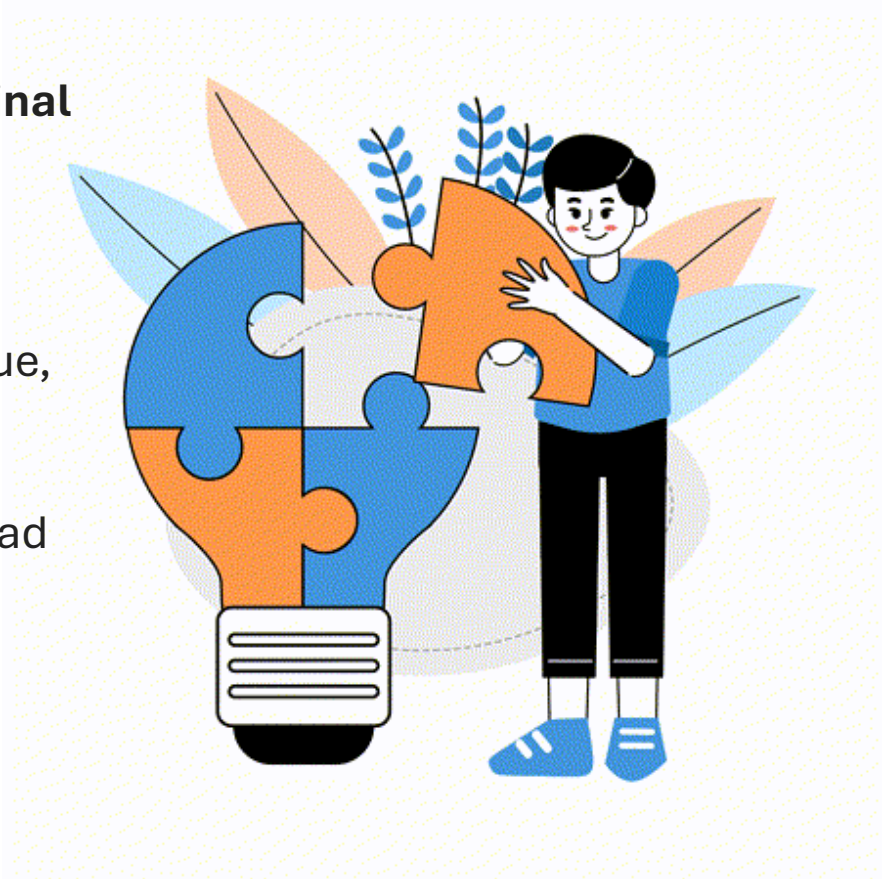
Problem Statement

The main problem in this project is: **How can we predict a student's final grade based on academic, family, and school-related information?**

The model tries to predict G3, which means the student's final grade.

This is a regression problem because the final grade is a numerical value, not a category.

For example, the model predicts a value such as 10, 12, 15, or 18 instead of predicting labels like pass or fail.



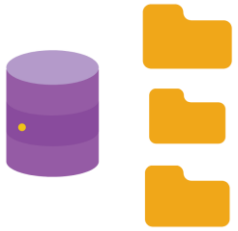
Dataset Description

The dataset includes two files. The first file is **student-mat.csv**, which contains student performance data for **Mathematics**.

The second file is **student-por.csv**, which contains student performance data for **Portuguese**.

The dataset includes many types of student information, such as demographic information, study time, absences, previous grades, family background, and school support.

The target variable is **G3**, which represents the final grade of the student.



student-mat.csv

Mathematics dataset
395 rows × 33 original columns

student-por.csv

Portuguese dataset
649 rows × 33 original columns

Combined data

1044 rows after combining
subject feature added

- Missing values: 0
- Duplicate rows: 0
- Final grade range: 0 to 20

Data detail in demo

Data Preprocessing

01 Load data

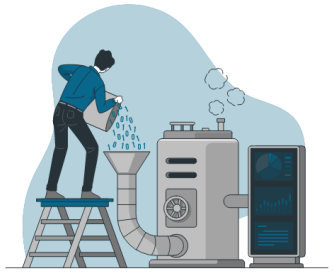
Read student-mat.csv and student-por.csv from the local data folder.

02 Combine files

Add subject column, then merge both datasets into one dataset.

03 Check quality

Inspect missing values, duplicates, data types, and descriptive statistics.

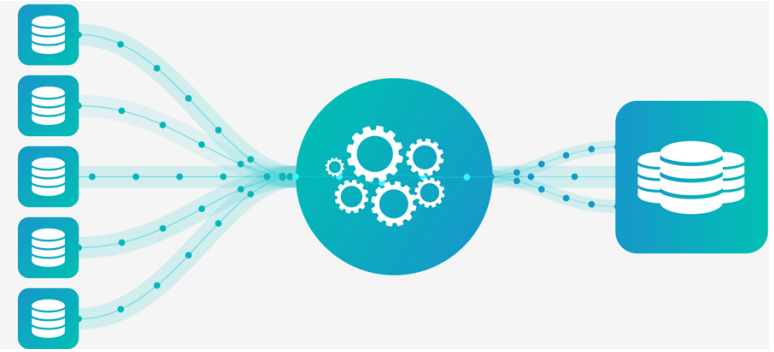


04 Encode categories

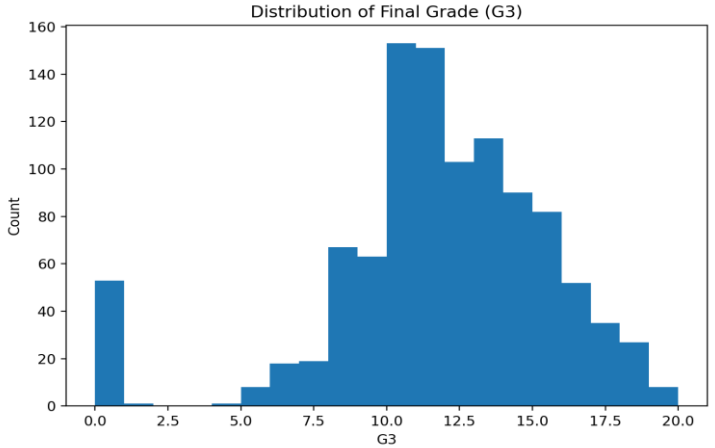
Convert categorical text features into numerical format.

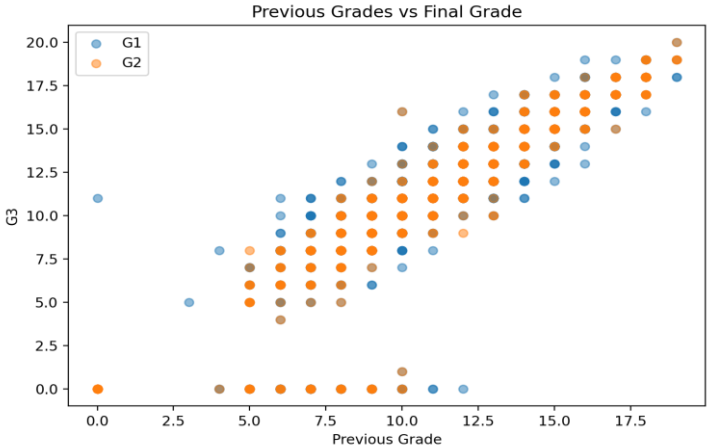
05 Split data

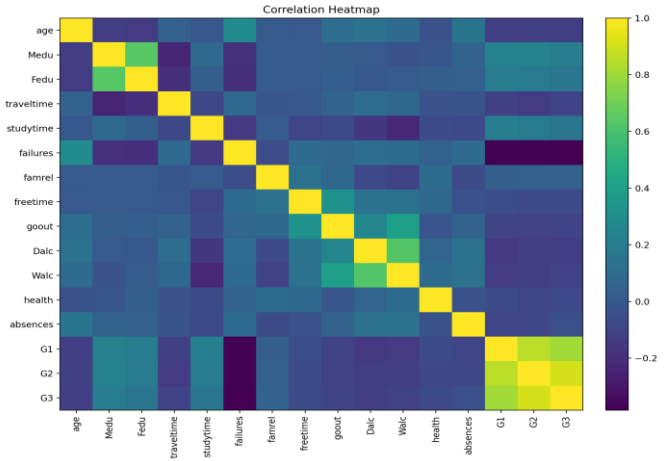
Separate X and y, then create train and test datasets.



Exploratory Data Analysis







- Distribution of final grade G3
- Correlation between numeric features
- Relationship between G1, G2, and G3
- Impact of absences, failures, and study time
- Categorical patterns such as school support and family support

Key Data Insights

01 Previous grades are strongest

G1 and G2 have strong relationship with G3, so past performance is a major predictor.

02 Failures reduce performance

Students with more previous failures usually have lower final grades.

03 Absences may matter

More absences can be connected with weaker academic performance.

04 Study time helps

Study time can influence outcomes, but it is not the only factor.

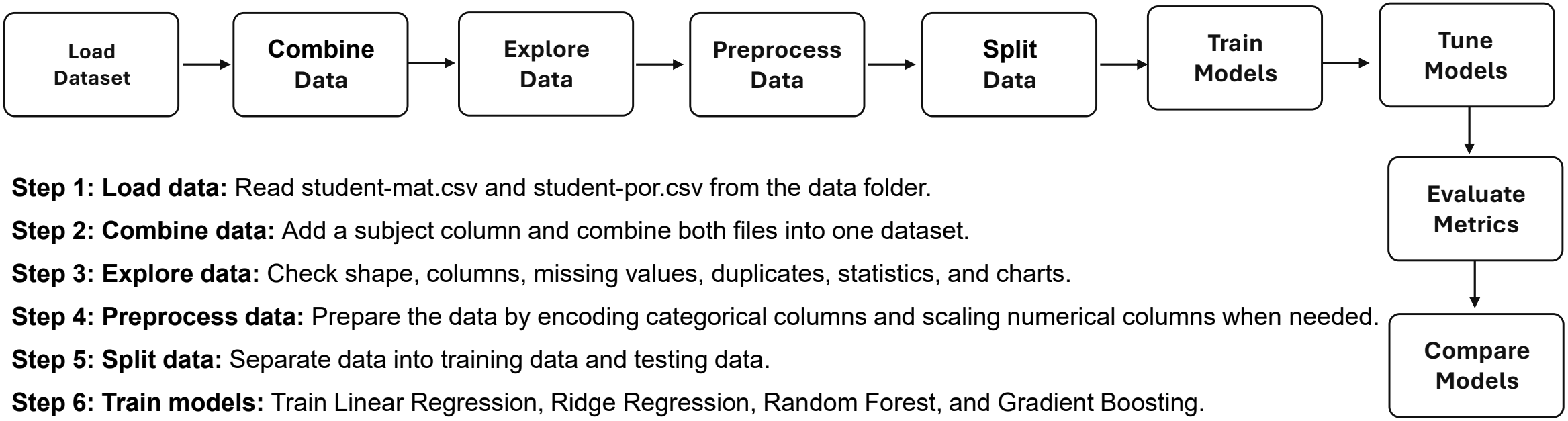
05 Family and support factors

School and family support features may also affect student outcomes.

06 Data is multifactorial

Student performance is affected by many academic and personal factors together.

Methodology



Step 1: Load data: Read student-mat.csv and student-por.csv from the data folder.

Step 2: Combine data: Add a subject column and combine both files into one dataset.

Step 3: Explore data: Check shape, columns, missing values, duplicates, statistics, and charts.

Step 4: Preprocess data: Prepare the data by encoding categorical columns and scaling numerical columns when needed.

Step 5: Split data: Separate data into training data and testing data.

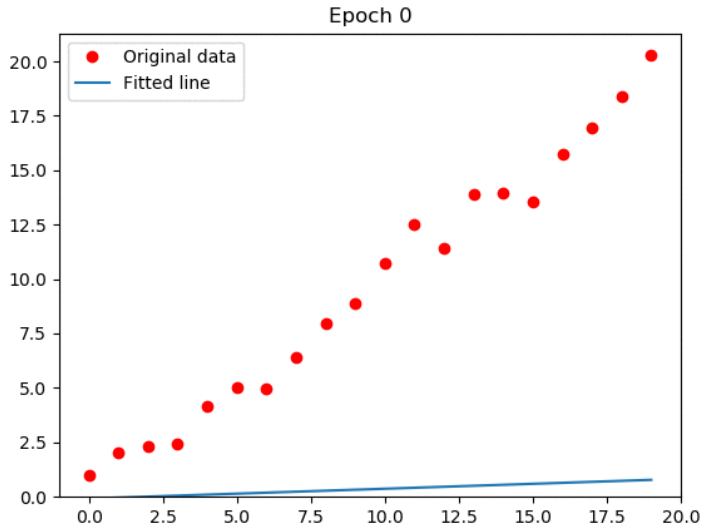
Step 6: Train models: Train Linear Regression, Ridge Regression, Random Forest, and Gradient Boosting.

Step 7: Tune models: Use GridSearchCV to find better model settings.

Step 8: Evaluate models: Compare RMSE, MAE, and R^2 on the test set.

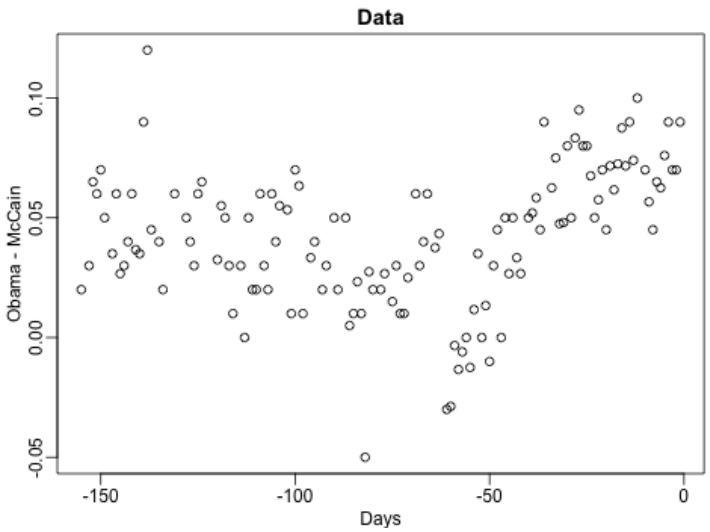
Step 9: Choose best model: Select the model with low error and high R^2.

Algorithms Used



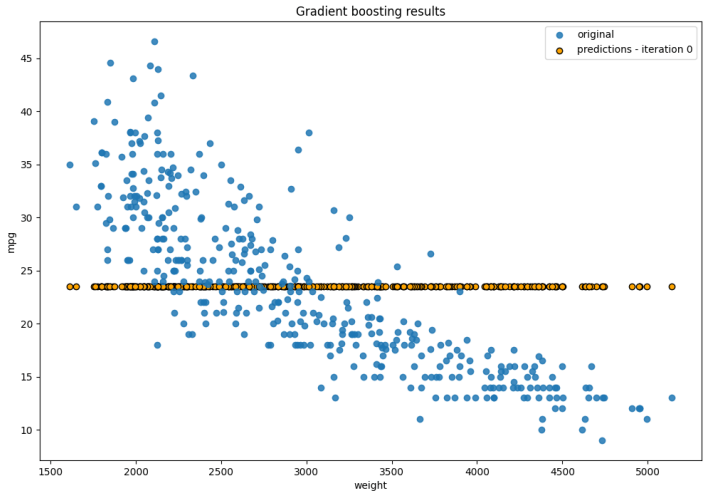
Linear Regression

A simple baseline model. It assumes a mostly linear relationship between features and final grade. It is easy to interpret.



Random Forest Regressor

An ensemble model that combines many decision trees. It can capture nonlinear relationships and feature interactions.



Gradient Boosting Regressor

A boosting model that improves step by step by learning from previous errors. It often performs well on tabular data.

All models were evaluated using the same train-test split and the same regression metrics

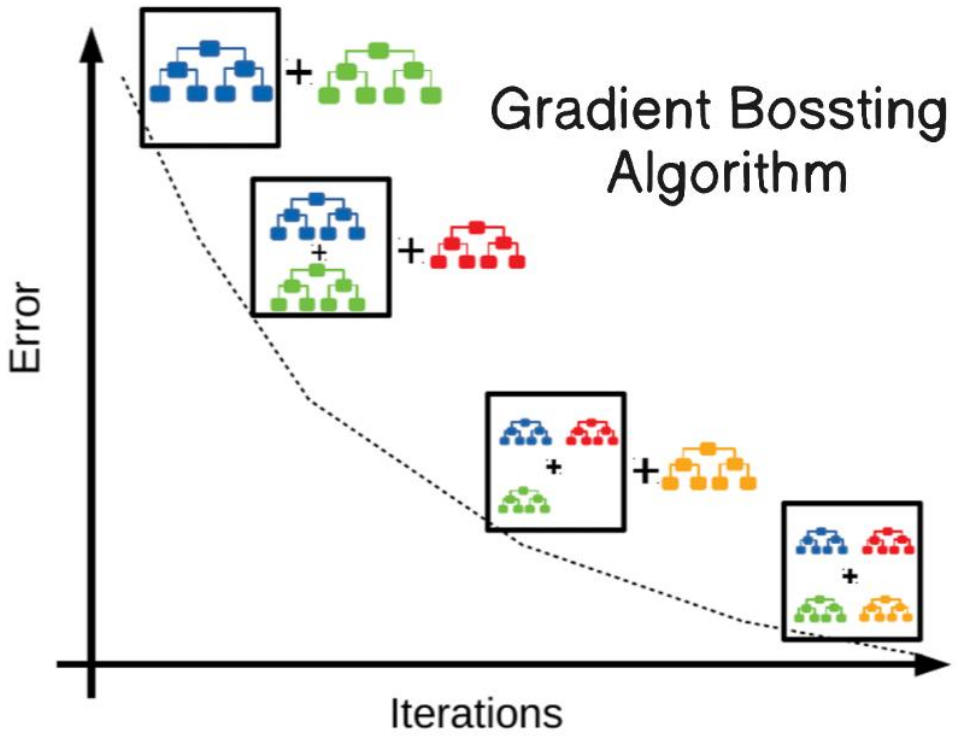
Linear Regression Algorithms

Random Forest Regressor Algorithms

Gradient Boosting Regressor Algorithm

Gradient Boosting Regressor is a supervised machine learning algorithm used for **regression problems**. It predicts continuous numerical values, such as student final grades.

In this project, Gradient Boosting is used to predict the final grade **G3** based on student information such as previous grades, study time, absences, failures, and family or school-related features.



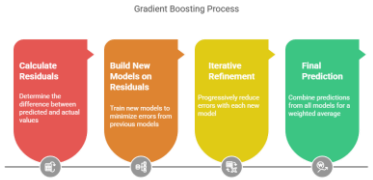
Gradient Boosting Regressor Algorithm

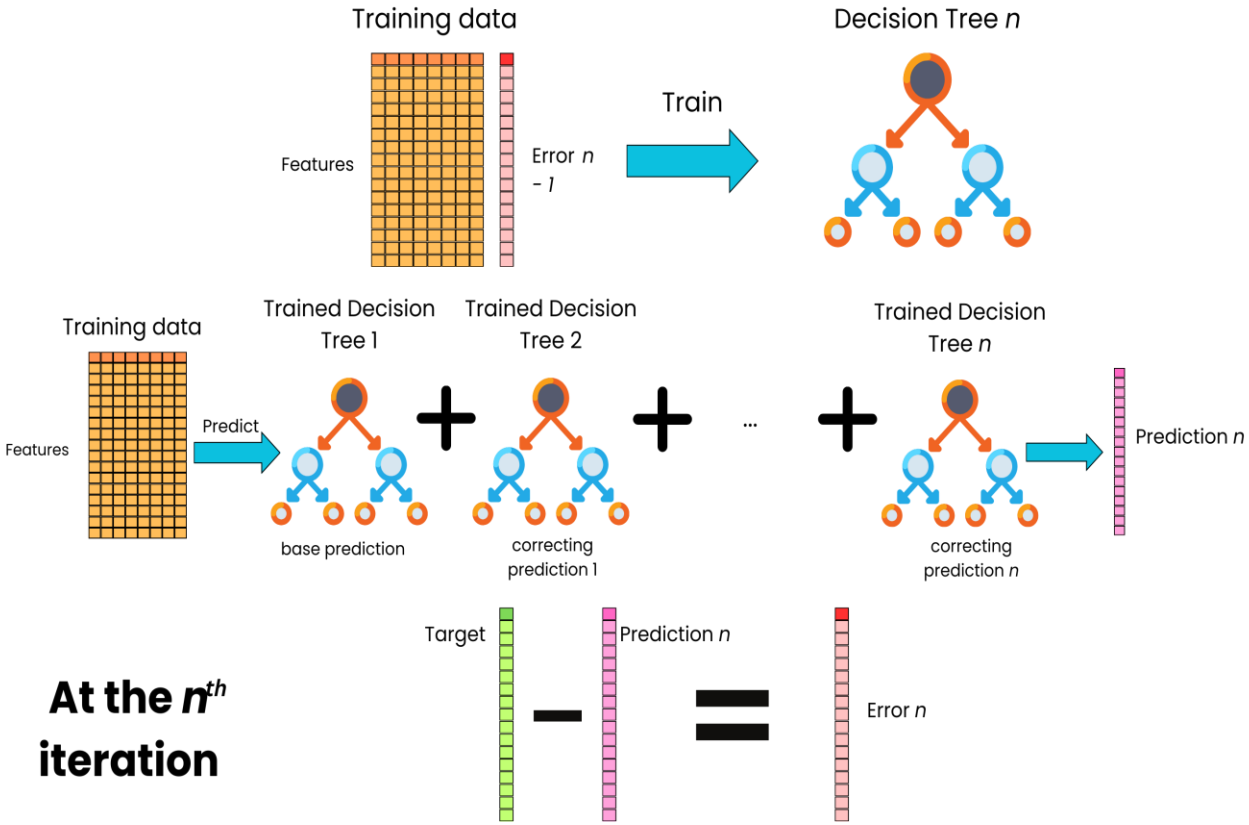
How Gradient Boosting Works

Gradient Boosting builds the model step by step.

1. It starts with a simple prediction.
2. It checks the prediction errors.
3. It trains a new small model to correct those errors.
4. It repeats this process many times.
5. The final prediction is made by combining all small models.

Each new model tries to improve the mistakes made by the previous models.





Gradient Boosting Regressor Algorithm

Why We Use Gradient Boosting

Gradient Boosting is useful because:

- It can handle complex relationships in the dataset.
- It works well with tabular data.
- It can improve prediction accuracy step by step.
- It is stronger than simple Linear Regression when the data is not fully linear.
- It can reduce prediction errors by learning from previous mistakes.

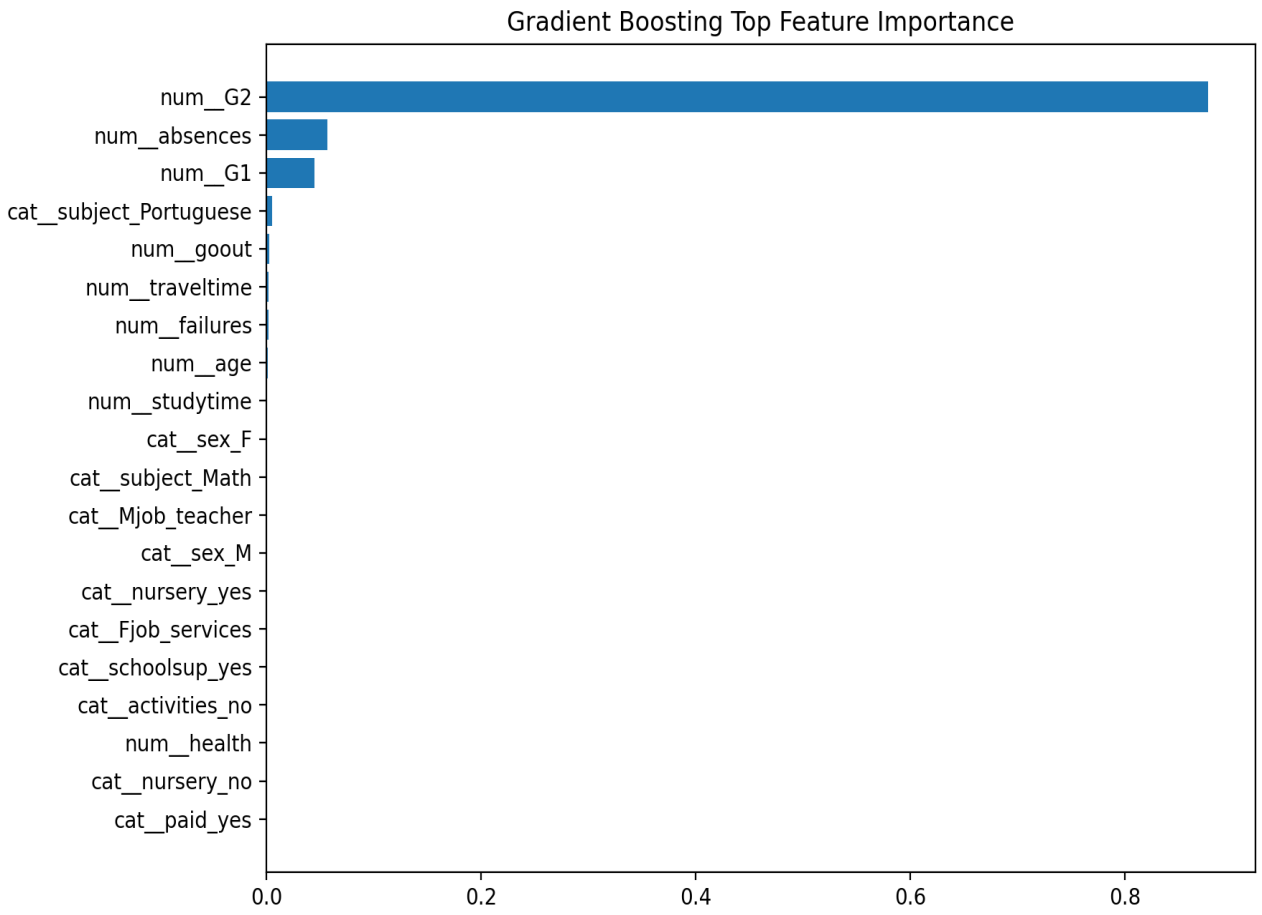
Gradient Boosting Regressor Algorithm

In This Project

For our Student Performance Prediction project:

- **Input features:** student demographic, academic, family, and school-related data
- **Target variable:** G3, the final grade
- **Task type:** Regression
- **Goal:** Predict a numerical final grade for each stud

Gradient Boosting performed well because student performance is affected by many factors, and the relationship between these factors is not always simp



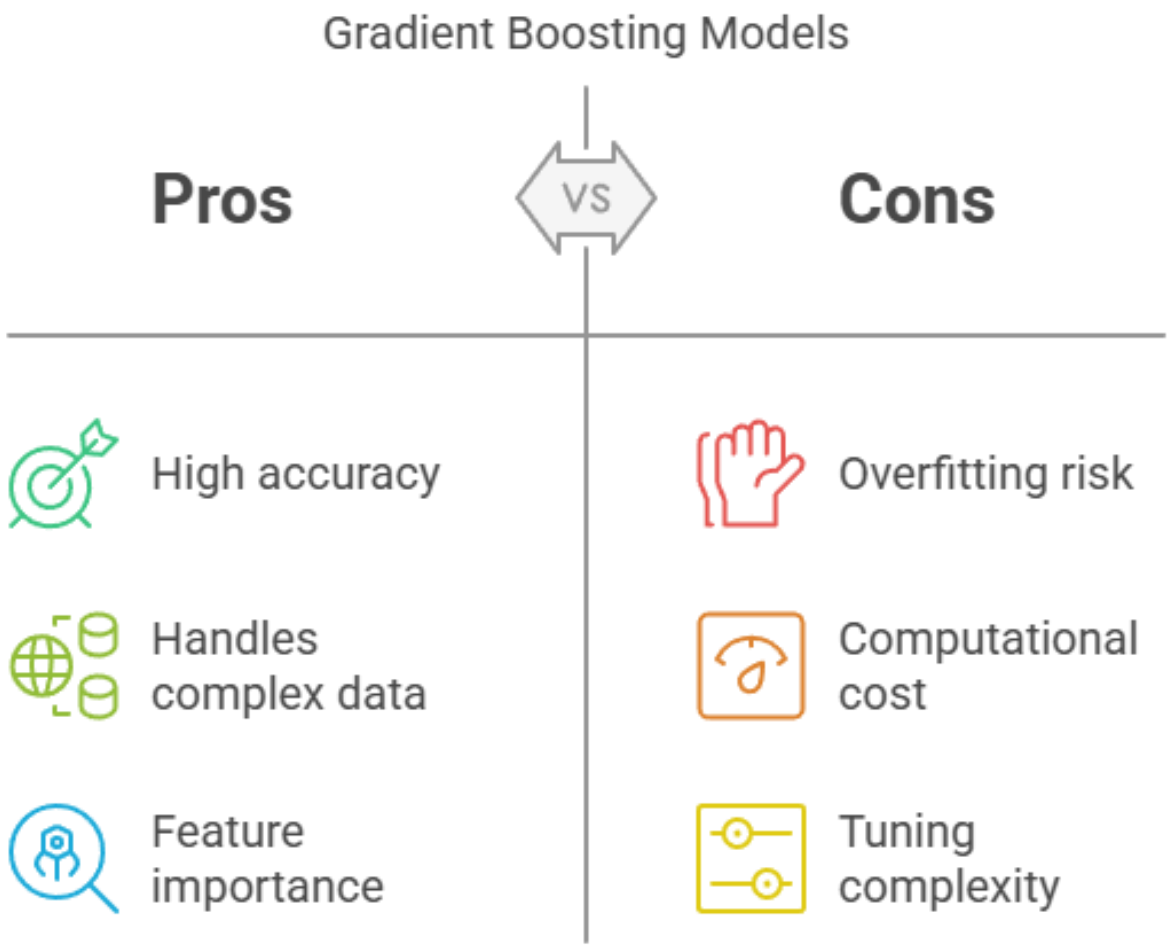
Gradient Boosting Regressor Algorithm

Strengths

- High prediction accuracy
- Good for nonlinear data
- Learns from previous errors
- Performs well on structured datasets
- Can capture feature interactions

Weaknesses

- More complex than Linear Regression
- Takes more time to train
- Needs careful hyperparameter tuning
- Can overfit if the model is too complex



Evaluation Metrics

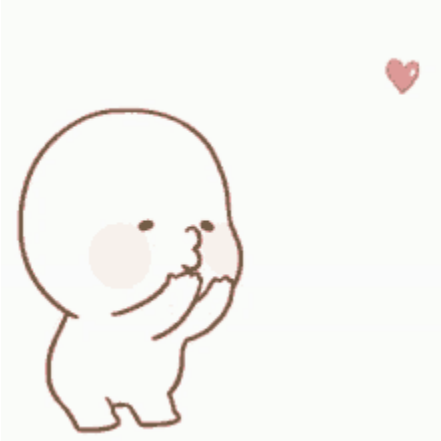
Model Comparison

Best Model Analysis

Challenges

Conclusion

Kong Sinh, Som Soknouch, Hoeurng Phally



Thank you very much for your attention and patience.
If you have any questions, please don’t hesitate to ask.